

Laplace transforms of integrals

1. Find Laplace transform of $\int_0^t e^{-t} \cos(3t) \cos(2t) dt$

ans: $\frac{F(s)}{s} = \frac{(s+1)(s^2+2s+14)}{s(s^2+2s+26)(s^2+2s+2)}$

2. Find Laplace transform of $\int_0^t te^{-t} \sin(2t) dt$

ans: $\frac{4(s+1)}{s(s^2+2s+5)^2}$